

ME 333
MECHANICS OF MATERIALS

Lecture 7: Stresses on an arbitrary plane &
Stress Transformations

Ravi Sastri Ayyagari

Mechanical Engineering
Indian Institute of Technology Gandhinagar

Sem I, 2025-2026

Stress Tensor

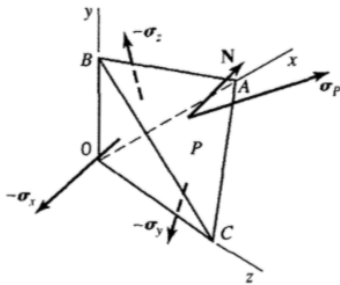
Q How is the traction vector related to the second order stress tensor?

Consider a traction vector acting on any arbitrary plane with a normal with direction cosines (l, m, n) . The components of the traction vector are linear combination of the direction cosines with respective components of the stress tensor. Let us prove it !

Stresses on an arbitrary plane

Stresses on an arbitrary plane

- Consider an arbitrary plane with a normal $N = l\hat{i} + m\hat{j} + n\hat{k}$.
- Consider a *Tetrahedral element*:



- Vector Area:

$$A = AN = A(l\hat{i} + m\hat{j} + n\hat{k}) \implies A_x = A \cdot \hat{i} = Al$$

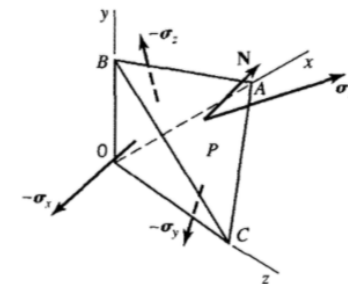
Ref Fig: A. P. Boresi and R. J. Schmidt, Advanced Mechanics of Materials, Sixth Edition.

Stresses on an arbitrary plane

Stresses on an arbitrary plane

- Traction vector can be written as:

$$\sigma_P = \sigma_{Px}\hat{i} + \sigma_{Py}\hat{j} + \sigma_{Pz}\hat{k}$$

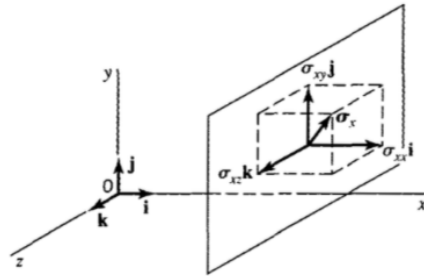


- Force equilibrium: $\sigma_P = l\sigma_x + m\sigma_y + n\sigma_z$

Ref Fig: A. P. Boresi and R. J. Schmidt, Advanced Mechanics of Materials, Sixth Edition.

Stresses on an arbitrary plane

- The Traction vector σ_x can be written as: $\sigma_x = \sigma_{xx}\hat{i} + \sigma_{xy}\hat{j} + \sigma_{xz}\hat{k}$



- Components of traction vector can therefore be written as:

$$\sigma_{Px} = l\sigma_{xx} + m\sigma_{yx} + n\sigma_{zx}; \quad \sigma_{Py} = l\sigma_{xy} + m\sigma_{yy} + n\sigma_{yz}; \quad \sigma_{Pz} = l\sigma_{xz} + m\sigma_{yz} + n\sigma_{zz}$$

Ref Fig: A. P. Boresi and R. J. Schmidt, Advanced Mechanics of Materials, Sixth Edition.

Normal and Shear stresses

- Normal stress can be written as:

$$\begin{aligned} \sigma_{PN} &= \sigma_P \cdot N = (\sigma_{Px}\hat{i} + \sigma_{Py}\hat{j} + \sigma_{Pz}\hat{k}) \cdot (l\hat{i} + m\hat{j} + n\hat{k}) \\ &= l\sigma_{Px} + m\sigma_{Py} + n\sigma_{Pz} \\ &= l(l\sigma_{xx} + m\sigma_{yx} + n\sigma_{zx}) + m(l\sigma_{xy} + m\sigma_{yy} + n\sigma_{yz}) + n(l\sigma_{xz} + m\sigma_{yz} + n\sigma_{zz}) \\ &= l^2\sigma_{xx} + m^2\sigma_{yy} + n^2\sigma_{zz} + 2lm\sigma_{yz} + 2ln\sigma_{xz} + 2lm\sigma_{xy} \end{aligned}$$

- σ_{PN} attains maximum value on one such arbitrary plane that corresponds to *Principal Plane* and the maximum normal stress on this plane is *Principal Stress*.
- The shear stress in the plane then is:

$$\sigma_{PS} = \sqrt{\sigma_P^2 - \sigma_{PN}^2} = \sqrt{\sigma_{Px}^2 + \sigma_{Py}^2 + \sigma_{Pz}^2 - \sigma_{PN}^2}$$

Stress Transformations

- Transformations use direction cosines:

	cos	x	y	z
x'	l ₁	m ₁	n ₁	
y'	l ₂	m ₂	n ₂	
z'	l ₃	m ₃	n ₃	

- Normal stress on any plane given the state of stress at a point is:

$$\sigma_{x'x'} = N_1 \cdot \sigma \cdot N_1 = l_1^2\sigma_{xx} + m_1^2\sigma_{yy} + n_1^2\sigma_{zz} + 2m_1n_1\sigma_{yz} + 2n_1l_1\sigma_{zx} + 2l_1m_1\sigma_{xy}$$

- Shear stress on any plane given the state of stress at a point is:

$$\begin{aligned} \sigma_{x'y'} &= N_1 \cdot \sigma \cdot N_2 = l_1l_2\sigma_{xx} + m_1m_2\sigma_{yy} + n_1n_2\sigma_{zz} + (m_1n_2 + m_2n_1)\sigma_{yz} \\ &\quad + (l_1n_2 + l_2n_1)\sigma_{zx} + (m_1l_2 + m_2l_1)\sigma_{xy} \end{aligned}$$

Stress Invariants

- Similar to strain Invariants, stress invariants can be determined by:

$$\sigma = \begin{vmatrix} \sigma_{xx} - \lambda & \sigma_{xy} & \sigma_{xz} \\ \sigma_{xy} & \sigma_{yy} - \lambda & \sigma_{yz} \\ \sigma_{xz} & \sigma_{yz} & \sigma_{zz} - \lambda \end{vmatrix} = 0$$

- The characteristic equation is:

$$\lambda^3 - I_1\lambda^2 + I_2\lambda - I_3 = 0$$

- Invariants of stress tensor are:

$$\begin{aligned} I_1 &= \sigma_{xx} + \sigma_{yy} + \sigma_{zz} = \sigma_1 + \sigma_2 + \sigma_3 = \text{tr}(\sigma) \\ I_2 &= \sigma_{xx}\sigma_{yy} + \sigma_{yy}\sigma_{zz} + \sigma_{zz}\sigma_{xx} - \sigma_{xy}^2 - \sigma_{yz}^2 - \sigma_{zx}^2 = \sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_3\sigma_1 \\ I_3 &= \sigma_1\sigma_2\sigma_3 = \det(\sigma) \end{aligned}$$

Mean and Deviatoric Stresses

- Mean stress can be defined as:

$$\sigma_m = \frac{1}{3}I_1 = \frac{\sigma_{xx} + \sigma_{yy} + \sigma_{zz}}{3} = \frac{\sigma_1 + \sigma_2 + \sigma_3}{3}$$

- Stress tensor can be additively decomposed into mean and deviatoric components.

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}_m + \boldsymbol{\sigma}_d$$

- Deviatoric and mean stress tensors can then be defined as:

$$\boldsymbol{\sigma}_d = \begin{bmatrix} \sigma_{xx} - \sigma_m & \sigma_{xy} & \sigma_{xz} \\ \sigma_{xy} & \sigma_{yy} - \sigma_m & \sigma_{yz} \\ \sigma_{xz} & \sigma_{yz} & \sigma_{zz} - \sigma_m \end{bmatrix}; \quad \boldsymbol{\sigma}_m = \begin{bmatrix} \sigma_m & 0 & 0 \\ 0 & \sigma_m & 0 \\ 0 & 0 & \sigma_m \end{bmatrix}$$

Characteristics of Deviatoric Stress Tensor

- $\boldsymbol{\sigma}_d$ is defined as:

$$\boldsymbol{\sigma}_d = \begin{bmatrix} \sigma_{xx} - \sigma_m & \sigma_{xy} & \sigma_{xz} \\ \sigma_{xy} & \sigma_{yy} - \sigma_m & \sigma_{yz} \\ \sigma_{xz} & \sigma_{yz} & \sigma_{zz} - \sigma_m \end{bmatrix}$$

- $\boldsymbol{\sigma}_d$ represents a state of pure shear. The principal values of deviatoric stress tensor are:

$$S_1 = \sigma_1 - \sigma_m = \frac{2\sigma_1 - \sigma_2 - \sigma_3}{3} = \frac{(\sigma_1 - \sigma_3) + (\sigma_1 - \sigma_2)}{3}$$

$$S_2 = \sigma_2 - \sigma_m = \frac{(\sigma_2 - \sigma_3) + (\sigma_2 - \sigma_1)}{3}$$

$$S_3 = \sigma_3 - \sigma_m = \frac{(\sigma_3 - \sigma_1) + (\sigma_3 - \sigma_2)}{3}$$

- The trace of $\boldsymbol{\sigma}_d$ is zero. Therefore, only two Principal stress values are independent.
- The Principal directions of $\boldsymbol{\sigma}_d$ are same as that of $\boldsymbol{\sigma}$

Reading Assignment

Chapter 2: Section 2.1 - 2.3.1